

Permutation tests

Used for hypothesis testing on measures of association

Generates a null distribution by mixing the real data randomly, multiple times

EXAMPLE 13.5: Sexual cannibalism in sagebrush crickets



The sage cricket, *Cyphoderris strepitans*, has an unusual form of mating. During mating, the male offers his fleshy hind wings to the female to eat. The wounds are not fatal,¹⁰ but a male with already nibbled wings is less likely to be chosen by females he meets later.



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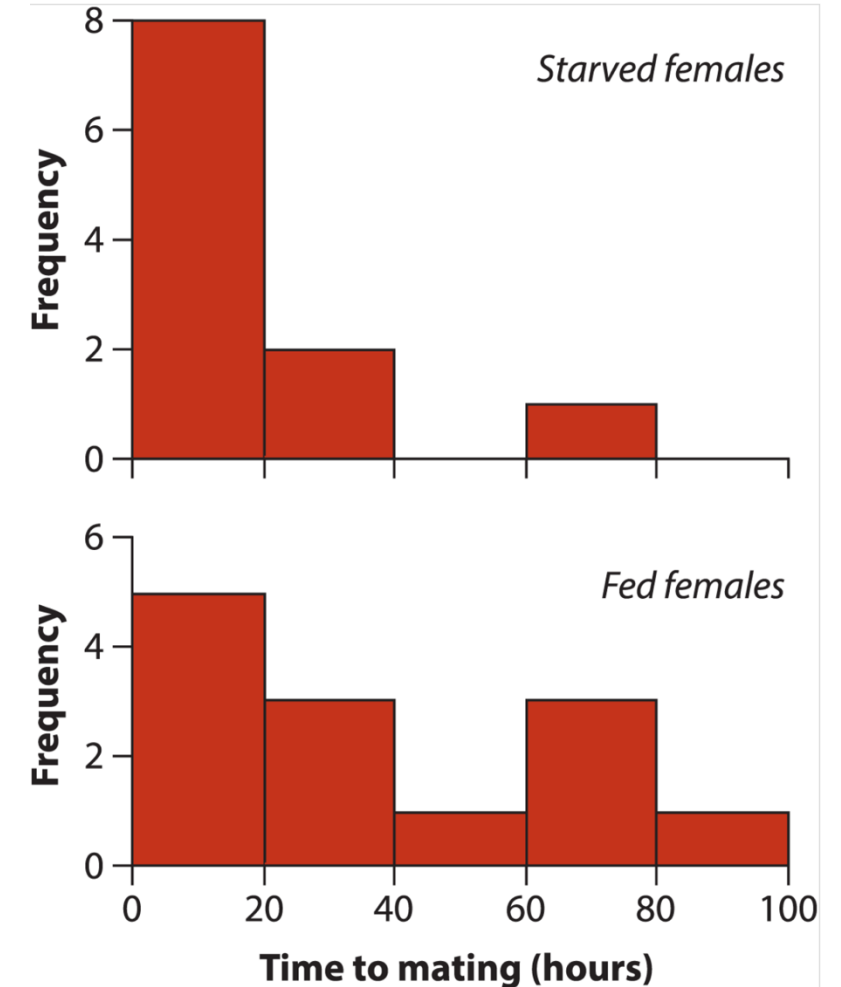
Females get some nutrition from feeding on the wings, which raises the question, “Are females more likely to mate if they are hungry?” Johnson et al. (1999) answered this question by randomly dividing 24 females into two groups: one group of 11 females was starved for at least two days and another group of 13 females was fed during the same period. Finally, each female was put separately into a cage with a single (new) male, and the waiting time to mating was recorded. The data are listed in [Table 13.5-1](#). The median time to mating was 13.0 hours for starved females and 22.8 hours for fed females.

H_0 : Mean time to mating is the same for female crickets that were starved and for those that were fed.

H_A : Mean time to mating differs between these groups.

TABLE 13.8-1 Times to mating (in hours) of female sagebrush crickets that were recently starved or fed. Data from the two treatments are color-coded to more easily identify the origin of each value later in [Table 13.8-2](#).

Treatment	Time (hours)	Treatment	Time (hours)
Starved	1.9	Fed	1.5
Starved	2.1	Fed	1.7
Starved	3.8	Fed	2.4
Starved	9.0	Fed	3.6
Starved	9.6	Fed	5.7
Starved	13.0	Fed	22.6
Starved	14.7	Fed	22.8
Starved	17.9	Fed	39.0
Starved	21.7	Fed	54.4
Starved	29.0	Fed	72.1
Starved	72.3	Fed	73.6
		Fed	79.5
		Fed	88.9
Mean	17.73	Mean	35.98



(log transform doesn't help here)

$$\bar{Y}_1 - \bar{Y}_2 = 17.73 - 35.98 = -18.26$$

The outcome of ONE permutation:

Treatment	Time (hrs)	Treatment	Time (hrs)
Starved	3.8	Fed	14.7
Starved	9.0	Fed	21.7
Starved	3.6	Fed	1.7
Starved	79.5	Fed	2.1
Starved	17.9	Fed	1.5
Starved	22.8	Fed	2.4
Starved	54.4	Fed	5.7
Starved	13.0	Fed	39.0
Starved	9.6	Fed	29.0
Starved	1.9	Fed	72.1
Starved	22.6	Fed	88.9
		Fed	72.3
		Fed	73.6
Mean	21.65	Mean	32.67

$$\bar{Y}_1 - \bar{Y}_2 = 21.65 - 32.67 = -11.02$$

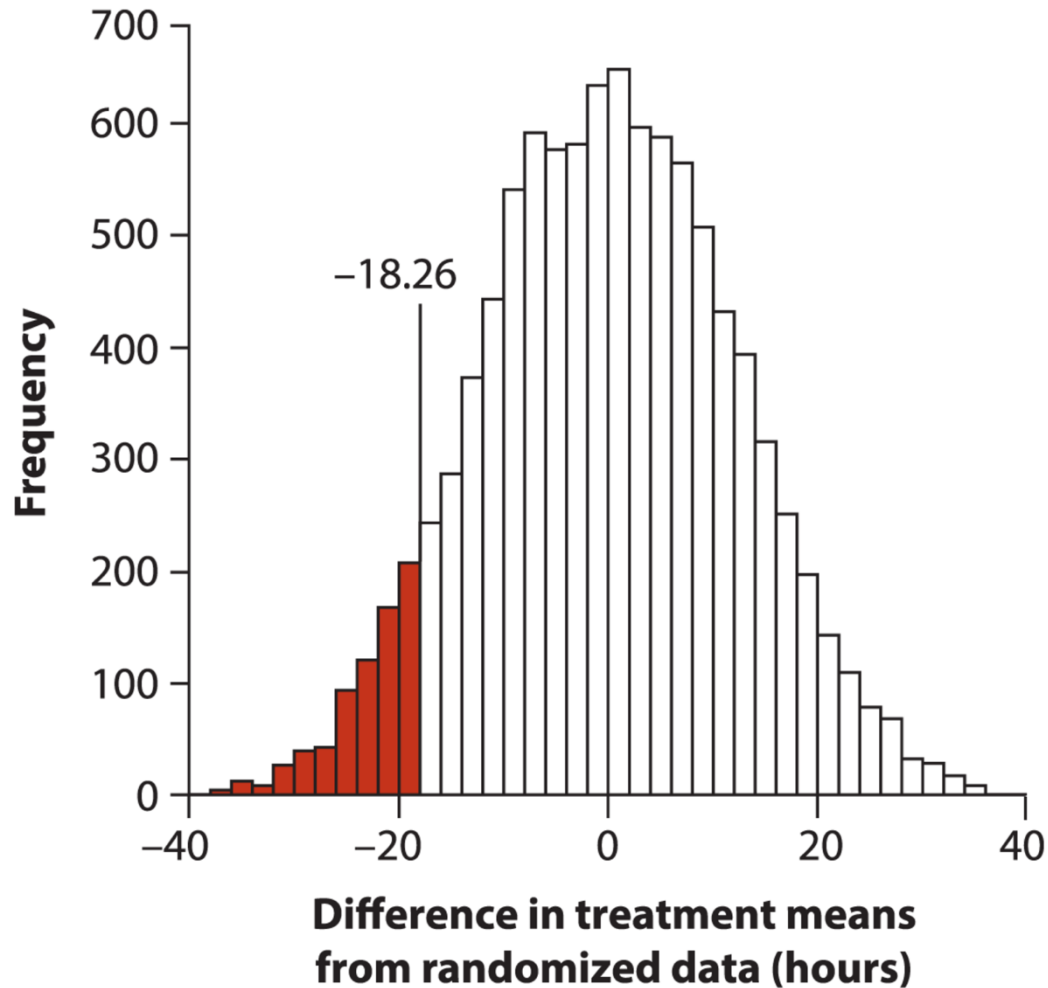
This is a plausible outcome under H_0

“Without replacement”

In other words, all data points are used exactly once in each permuted data set.

We repeat this process thousands of times more, each time computing $\bar{Y}_1 - \bar{Y}_2$

The outcome of 10000 permutations



The histogram comprises values of $\bar{Y}_1 - \bar{Y}_2$ from each sampled permutation.

It forms our null distribution.

We see that 712 of the 10000 permutations had a $\bar{Y}_1 - \bar{Y}_2$ lower than -18.26 (*observed $\bar{Y}_1 - \bar{Y}_2$ from the study*).

To get a two-tailed P-value:

$$2(0.0712) = 0.142$$

We therefore fail to reject the null. We are unable to conclude that starved and fed females differ in mean time to mating.